RESAD SPAHOVIC

Python Engineer | AI Systems Engineer | Backend, Data & ML Infrastructure

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PROFESSIONAL SUMMARY

Python Engineer with 4+ years of experience across AI systems, backend development, data engineering and ML infrastructure. For the past two years I have designed and written a complete clinic management system — scheduling, clinical records, laboratory, operating theatre, home visits, stock and finance in one product — and now run it as my own company, MAAT INNOVATIONS. I am used to owning a system end to end: database schema and migrations, API, authentication and roles, encryption of personal data, document generation, integrations, and the tests that keep all of it honest.

CORE TECHNICAL STACK

Programming & Backend	Python, SQL, JavaScript, FastAPI, Django, Flask, REST APIs, SQLAlchemy, Alembic, production backend architecture
AI / Machine Learning	TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, NLP, Computer Vision, Time-Series Forecasting
LLM / Retrieval	RAG pipelines, embeddings, vector search, pgvector, Qdrant, chatbot-assisted support, trusted retrieval workflows
Databases & Storage	PostgreSQL, MongoDB, Redis, DuckDB, MySQL, SQLite, Qdrant
Cloud & DevOps	AWS, GCP, Docker, Kubernetes, CI/CD, Prometheus metrics, health/readiness endpoints
Security & Data	JWT, RBAC, 2FA (TOTP), CSRF, encryption at rest, blind indexes, ETL pipelines, audit logs, rate limiting, automated testing

PROFESSIONAL EXPERIENCE

MAAT INNOVATIONS - Founder & Software Engineer, Novi Pazar — 2025 - Present

- Design and write MAAT Clinic, a system that covers the whole provider rather than one department: scheduling, clinical records, laboratory (LIS), operating theatre, home visits, stock, equipment and finance.
- Own every layer: PostgreSQL schema and Alembic migrations, FastAPI services, role-based access for eight roles, encryption of personal and medical fields at rest with blind-index search, PDF reports, e-mail and SMS.
- Integrations where they matter: laboratory analysers over ASTM, FHIR R4 export, fiscal receipts (ESIR / eFaktura).
- Every defect found becomes a regression test that must fail on the old code before the fix lands; the suite runs in CI.

LINK Group - Python Engineer, Belgrade — 2024 - Present

- Architected and deployed ML-powered applications serving 50,000+ daily active users.
- Built a predictive analytics platform for client decision support: forecasting, anomaly detection and dashboard reporting.
- Rewrote data processing pipelines, cutting a nightly run from 4 hours to 45 minutes.
- Mentored 5 junior developers; introduced automated testing in CI and a review checklist for ML model deployment.

Center NIT - Python Programming and Machine Learning Instructor — 2022 - 2023

- Designed and delivered the Python and Machine Learning curriculum for 200+ students.
- Built hands-on projects modelled on real industry work across Python, ML and data workflows.
- Created an automated assessment tool that took routine grading off the instructors.

City of Novi Pazar - System Administration — 2019 - 2022

- Managed IT infrastructure for 500+ employees across multiple government departments.
- Implemented automated backups with a restore procedure that was rehearsed, not assumed.
- Developed internal Python tools to automate routine administrative workflows.
- Took part in digital transformation of citizen-facing services.

Private Instructor - Mathematics and Computer Science, Belgrade — 2017 - 2019

- Provided personalized instruction to 100+ students in advanced mathematics and programming.
- Developed custom learning materials and interactive coding exercises.

SELECTED PROJECTS

MAAT Clinic - Clinic Management System (own product)

- One system for the whole clinic: online and phone booking, clinical records and reports, laboratory orders and results, operating theatre with mandatory anaesthetist clearance before admission, home visits with technician dispatch and stock issue, inventory with recipe-based consumption, equipment service and calibration that blocks an overdue instrument, and finance down to profit per department.
- Billing as it really happens: part cash, part card, partial payments carried across visits, discounts and referrers, with every payment recording who took it, when and how much.
- Patient data encrypted at rest and searched through blind indexes rather than plaintext; every read of a medical record is written to an append-only audit trail.
- In-house RAG assistant (local embeddings in pgvector, no external LLM API) that answers patient questions and explains a report in plain language.

Key technologies: FastAPI, PostgreSQL, pgvector, SQLAlchemy, Alembic, Docker, ReportLab, JWT, RBAC, HTML, CSS, JavaScript

Automated Predictive Analytics System for Business Intelligence

- AI-powered platform for historical data analysis, trend forecasting, anomaly detection and customer behaviour analysis.
- ETL workflows with Pandas, NumPy and Apache Spark across several databases, with dashboard reporting in Dash/Plotly.

Key technologies: FastAPI, TensorFlow, PyTorch, Pandas, NumPy, Apache Spark, PostgreSQL, MongoDB, DuckDB, Dash/Plotly

Datacenter Management API

- REST API for datacenter devices and racks with rack-unit capacity validation, power consumption checks and balanced placement proposals.
- JWT authentication, RBAC, audit logging, health/readiness endpoints, Prometheus metrics, rate limiting, optimistic locking and idempotent writes.

Key technologies: FastAPI, SQLAlchemy, SQLite/PostgreSQL, Docker, JWT, RBAC, Prometheus metrics, audit logs

AI Legal Assistant Platform

- Legal support platform built around document understanding, trusted retrieval and guided assistance workflows, with answers tied back to the source document.

Key technologies: RAG, vector search, Qdrant, backend services, legal document processing

Drug Interaction Analysis Using Artificial Intelligence

- System that identifies potential drug interactions across combined medication use and flags likely adverse effects.
- Machine learning and NLP over medical text for risk assessment and safer recommendation workflows.

Key technologies: Flask, FastAPI, Scikit-learn, TensorFlow, PostgreSQL, spaCy, NLTK

Early Detection of Grapevine Diseases

- Deep learning over leaf imagery to catch subtle changes before a disease is visible to the grower, with stored history for season-long monitoring.

Key technologies: PyTorch, NumPy, Matplotlib, MySQL, Flask

EDUCATION

Master's Degree in Computer Science and Mathematics - State University of Novi Pazar	**2022 - 2024**
AI & Python Development - ITAcademy by LINK Group, Belgrade	**2023 - 2024**
HTML, CSS, JavaScript & React - Center NIT, Novi Pazar	**2022 - 2023**
State Professional Exam - Ministry of Public Administration and Local Self-Government	**2021**
Bachelor's Degree in Computer Science and Mathematics Education - State University of Novi Pazar	**2012 - 2017**
Computer and Electrical Technician - Technical School of Novi Pazar	**2008 - 2012**

CERTIFICATIONS

AI & Python Development - IT Academy by LINK Group	**2023 - 2024**
Full-Stack Web Development - Center NIT	**2022 - 2023**